

IncaDigital - DEX-CEX Market Analysis Challenge

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1 Task 1 — Hedged LP on Uniswap

We consider a liquidity position in an ETH/USDT pool, split equally by USD value (50% ETH and 50% USDT). Let the total portfolio value be V (in USD). The ETH price exposure of the portfolio is therefore $\frac{V}{2}$ USD. Let entry price of Ethereum is S_0 .

1. **Calculate the amount of ETH to short on a CEX perpetual to make the LP position locally delta-neutral at entry.**

Given that the portfolio is exposed to $\frac{V}{2} \times S_0$ amount of ETH, its initial delta should be :

$$\Delta_p = +\frac{VS_0}{2} ETH$$

To achieve a delta neutral strategy, we should short the same amount in a perp:

$$\Delta_0 = +\frac{VS_0}{2} - \frac{VS_0}{2} = 0ETH$$

2. **What other costs should be accounted for?**

- (a) **Funding rate** The perp position implies funding payments according to intervals, at $t=k$ with r as the funding rate, Q is the amount of ETH in the perp contract and P is the price of ETH :

$$Funding_k = r_k Q P_k$$

- (b) **Rebalancing fees on CEX to keep the delta neutral strat:** In this strat, as ETH price moves, we will need to rebalance on the perp

contract to match the amount ETH in the LP position to keep the delta neutral:

$$\text{ETH}_{LP} = \sqrt{\frac{k}{P}}$$

$$\Delta\text{ETH} = \text{ETH}_{\text{PERP}} - \text{ETH}_{LP}$$

$$\text{RebalancingFees}_{\text{USD}} = \Delta\text{ETH} \cdot \text{Price} \cdot \text{CEX_fee_rate}$$

Note: The rebalance transaction can be either a buy or a sell of ETH, depending on whether the position is underhedged or overhedged. Higher rebalance frequency increases transaction costs.

- (c) **Impermenant loss (IL):** IL comes from comparing the current strat to holding the ETH and simply doing nothing:

$$\text{IL} = \frac{2\sqrt{r}}{1+r} - 1, \quad r = \frac{P_2}{P_1}$$

so above is all the main costs I found for this start, and the daily PnL formula would be:

$$\text{DailyP\&L} = \text{Fees}_{LP} - \text{RealizedFundingPayment} - \text{IL} - \text{RebalancingFees}_{\text{USD}}$$

3. **How would your result change if, instead of V2, liquidity was provided in a Uniswap V3 pool within a $\pm 10\%$ range around the current price?**

Everything would stay the same, except :

- (a) **The amount of ETH to hedge locally, it will be less.**
- (b) **IL will be worst due to concentration of liquidity.**
- (c) **Perp will be closed when price is out of range for a conservative approach.**

for Uniswap V3, this strat could earn more fees but as a tradeoff, the IL will be higher than Uniswap V2.